

HANYU CHENG

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EDUCATION

Tsinghua University, Department of Industrial Engineering Sep 2021 – Jul 2026
Bachelor of Engineering, GPA: 3.8/4.0
Thesis: Optimizing Urban Electric Vehicle (EV) Battery Swapping and Charging Systems
Supervisor: Prof. Wei Qi

RESEARCH APPOINTMENTS

Staff Associate I, Columbia University Nov 2026 – Jan 2027
Supervisor: Prof. David D. Yao

Visiting Student Intern (VSI), Columbia University Mar 2026 – May 2026
Supervisor: Prof. David D. Yao

Visiting Student Researcher (VSR), University of California, Berkeley Jul 2025 – Sep 2025
Supervisor: Prof. Chiwei Yan

RESEARCH INTERESTS

Stochastic Control, Game Theory, Logistics.

WORKING PAPERS

Cheng H., & Yao, D. D. (2026). Optimal Control of a Non-Stationary Inventory-Queue, with Applications to EV-Battery Swapping. <https://arxiv.org/abs/2609.35153>

Gao, S., **Cheng, H.**, Yan, C., & Jiang, G. (2026). A Markovian Traffic Equilibrium Model for Ride-Hailing. Major Revision at *Transportation Science*. <https://doi.org/10.48550/arXiv.2604.21359>

PUBLICATIONS

Cheng, H., Cheng, L., & Bai, X. (2026). Dynamic Carbon Intensity Indicator (CII) Management in Stochastic Tramp Shipping Market. *Transportation Research Part A: Policy and Practice*, 210, 105059. <https://doi.org/10.1016/j.tra.2026.105059>

Ai, W.^{*}, **Cheng, H.**^{*}, & Qi, W. (2026). Optimizing Urban Electric Vehicle Charging and Battery Swapping Infrastructure: A Location-Inventory-Grid Model. *Transportation Research Part E: Logistics and Transportation Review*, 214, 104897. <https://doi.org/10.1016/j.tre.2026.104897>

Cheng, H., Tian, Q., & Wang, D.Z. (2025). Adding Runs of Modular Autonomous Vehicles with Skip-Stop Strategy Using Deep Reinforcement Learning. *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 4066-4073. <https://doi.org/10.1109/ITSC60802.2025.11423421>

^{*}These authors contributed equally.

REFEREING ACTIVITIES

IEEE International Conference on Intelligent Transportation Systems (ITSC), IEEE Intelligent Vehicles Symposium (IV).

AWARDS AND SCHOLARSHIPS

Innovation Excellence Scholarship, Tsinghua University	2024,2025
Comprehensive Excellence Scholarship, Tsinghua University	2022,2023
Outstanding Student Leader, Tsinghua University	2022